VEHICLE PARKING SYSTEM
TEST PLAN & TEST CASE DOCUMENT

SRS-aligned version based on FR-01 to FR-14 and NFR-01 to NFR-08

1. Test Plan Overview

Item	Details
Project	Vehicle Parking System
SRS Basis	Vehicle Parking System SRS containing FR-01 to FR-14 and NFR-01 to NFR-08.
Objective	Verify that every stated functional requirement works correctly for valid and invalid inputs and that the integrated system satisfies the specified non-functional requirements.
Testing Approach	Manual testing using Unit, Integration and System test cases, following the supplied test-case template.
Functional Coverage	FR-01 through FR-14.
Non-Functional Coverage	NFR-01 through NFR-08.
Test Environment	Vehicle Parking System application, configured database/server, supported web browser and test accounts/data.
Entry Criteria	Implemented build is available; required database and services are configured; tester can access the application and test accounts.
Exit Criteria	All planned cases have been executed; actual results are recorded; failures/defects are logged and re-tested where applicable.
Execution Note	Actual Result and Test Result are intentionally left for manual execution. A test is Pass only when actual behavior matches the expected result.

2. SRS Functional Requirements Traceability

SRS ID	Requirement	Priority
FR-01	User Registration/Login	High
FR-02	Vehicle Registration	High
FR-03	Parking Slot Management	High
FR-04	Slot Availability	High
FR-05	Vehicle Entry	High
FR-06	Vehicle Exit	High
FR-07	Parking Reservation	Medium
FR-08	Parking Fee Calculation	High
FR-09	Payment Management	High
FR-10	Parking History	Medium
FR-11	Notifications	Medium
FR-12	Admin Dashboard	High
FR-13	Search Vehicle	Medium
FR-14	Reports	Medium

3. Non-Functional Requirements Test Coverage

SRS ID	Requirement	Priority
NFR-01	Performance	High
NFR-02	Security	High
NFR-03	Reliability	High
NFR-04	Usability	Medium
NFR-05	Availability	High
NFR-06	Scalability	Medium
NFR-07	Maintainability	Medium
NFR-08	Data Integrity	High

4. Functional Test Cases

Each functional requirement below contains 8 test cases, combining Unit, Integration and System testing. These cases are aligned directly to the SRS wording. The supplied template's Actual Result and Test Result fields remain blank for manual execution.

FR-01 - User Registration/Login

SRS requirement: The system shall allow users and parking administrators to register and log in securely.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01_01	User Registration/Login	Validate user registration with valid details	Registration page is accessible	Enter valid name, email and password; submit.	Name: Ravi; Email: ravi@test.com; Password: Park@123	Account should be created successfully.	To be entered after manual execution	To be entered
UT_01_02	User Registration/Login	Reject registration with invalid email	Registration page is accessible	Enter an invalid email and submit.	Email: ravi@	System should reject the input and show validation.	To be entered after manual execution	To be entered
UT_01_03	User Registration/Login	Reject registration with missing required fields	Registration page is accessible	Leave required fields blank and submit.	Name: blank; Email: blank; Password: blank	Required-field messages should be displayed.	To be entered after manual execution	To be entered
IT_01_01	User Registration/Login	Verify registered credentials can be used to log in	A new account can be created	Register a user, then log in with the same credentials.	ravi@test.com / Park@123	Registration data should be persisted and login should succeed.	To be entered after manual execution	To be entered
IT_01_02	User Registration/Login	Prevent duplicate user registration	User account already exists	Register another account using the same email.	Email: ravi@test.com	Duplicate registration should be blocked.	To be entered after manual execution	To be entered
IT_01_03	User Registration/Login	Verify user session reaches correct dashboard	Valid user account exists	Log in and observe the post-login page.	Valid user credentials	User should reach the user dashboard with an authenticated session.	To be entered after manual execution	To be entered
ST_01_01	User Registration/Login	Verify complete user registration flow	Application is running	Register a new user from start to finish.	Anita; anita@test.com; Park@123	Registration should complete and the user should be able to access the system.	To be entered after manual execution	To be entered
ST_01_02	User Registration/Login	Verify failed login cannot access protected functions	Valid account exists	Enter correct email with wrong password and try to continue.	anita@test.com / Wrong@123	Authentication should fail and protected pages should remain inaccessible.	To be entered after manual execution	To be entered

FR-02 - Vehicle Registration

SRS requirement: The system shall allow users to add and manage vehicle details such as vehicle number, type, and owner information.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_02_01	Vehicle Registration	Add vehicle with valid details	User is logged in	Open vehicle registration; enter vehicle number, type and owner; save.	KA01AB1234; Car; Ravi	Vehicle should be saved successfully.	To be entered after manual execution	To be entered
UT_02_02	Vehicle Registration	Reject invalid vehicle number format	User is logged in	Enter an invalid vehicle number and save.	ABC123	Invalid vehicle number should be rejected.	To be entered after manual execution	To be entered
UT_02_03	Vehicle Registration	Reject missing vehicle information	User is logged in	Leave required vehicle field blank and save.	Vehicle Number: blank; Type: Car	Required validation should be shown.	To be entered after manual execution	To be entered
IT_02_01	Vehicle Registration	Verify saved vehicle is linked to the correct user	User is logged in; database is available	Add a vehicle, log out, log in again and view vehicles.	KA02CD5678; SUV	Vehicle should appear under the same user's records.	To be entered after manual execution	To be entered
IT_02_02	Vehicle Registration	Prevent duplicate vehicle number for same user	Vehicle already exists	Attempt to add the same vehicle number again.	KA01AB1234	Duplicate vehicle should be rejected or handled per system rule.	To be entered after manual execution	To be entered
IT_02_03	Vehicle Registration	Update vehicle details	Existing vehicle is available	Edit vehicle type/owner information and save.	KA01AB1234; Type: SUV	Updated details should be stored and displayed.	To be entered after manual execution	To be entered
ST_02_01	Vehicle Registration	Delete a registered vehicle	Existing vehicle is available	Select vehicle, choose delete and confirm.	KA01AB1234	Vehicle should be removed from the user's active vehicle list.	To be entered after manual execution	To be entered
ST_02_02	Vehicle Registration	Use registered vehicle in a parking transaction	User and vehicle exist	Start a parking flow and select the registered vehicle.	KA02CD5678	The selected registered vehicle should be accepted by the parking flow.	To be entered after manual execution	To be entered

FR-03 - Parking Slot Management

SRS requirement: The system shall allow administrators to add, update, and remove parking slots.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_03_01	Parking Slot Management	Add a parking slot	Administrator is logged in	Open slot management; enter valid slot details; save.	Slot A01; Ground Floor	Slot A01 should be created.	To be entered after manual execution	To be entered
UT_03_02	Parking Slot Management	Reject duplicate slot ID	Slot A01 already exists	Enter A01 again and save.	Slot A01	Duplicate slot creation should be blocked.	To be entered after manual execution	To be entered
UT_03_03	Parking Slot Management	Update slot details/status	Administrator is logged in; slot exists	Edit slot data/status and save.	A01 -> Maintenance	Updated slot information should be displayed.	To be entered after manual execution	To be entered
UT_03_04	Parking Slot Management	Remove a parking slot	Administrator is logged in; removable slot exists	Select slot; choose remove; confirm.	Slot A02	Slot should be removed according to system rules.	To be entered after manual execution	To be entered
IT_03_01	Parking Slot Management	Verify added slot appears in availability data	Admin can add slots; availability module exists	Add A03 and open availability view.	Slot A03	A03 should appear with the correct state.	To be entered after manual execution	To be entered
IT_03_02	Parking Slot Management	Verify updated slot status affects booking	Admin and booking modules are available	Mark A03 unavailable; attempt user booking.	A03; Status: Unavailable	User should not be able to book A03.	To be entered after manual execution	To be entered
ST_03_01	Parking Slot Management	Manage multiple slots through admin interface	Administrator is logged in	Add, update and remove representative slots.	A04, A05	Admin actions should consistently update the parking inventory.	To be entered after manual execution	To be entered
ST_03_02	Parking Slot Management	Prevent unauthorized slot management	Non-admin user is logged in	Attempt to access slot management directly.	Normal user session	User should be denied admin-only slot management.	To be entered after manual execution	To be entered

FR-04 - Slot Availability

SRS requirement: The system shall display currently available and occupied parking slots.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_04_01	Slot Availability	Display available slots	Slot inventory is configured	Open the slot availability view.	A01 available; A02 occupied	Available and occupied statuses should be shown correctly.	To be entered after manual execution	To be entered
UT_04_02	Slot Availability	Display occupied slots correctly	At least one slot is occupied	Open availability view.	A02 occupied	A02 should be shown as occupied.	To be entered after manual execution	To be entered
UT_04_03	Slot Availability	Refresh availability after a status change	Slot status can be changed	Refresh/reload availability after changing a slot.	A03 changes Available -> Occupied	Displayed status should reflect the latest state.	To be entered after manual execution	To be entered
IT_04_01	Slot Availability	Reflect admin slot status in availability	Admin can update slots	Admin changes a slot to maintenance/unavailable; user views availability.	A04 -> Maintenance	A04 should not be shown as normally available.	To be entered after manual execution	To be entered
IT_04_02	Slot Availability	Reflect vehicle entry allocation	Entry module is available	Record an entry that allocates an available slot; open availability.	Vehicle: KA01AB1234; Slot A01	A01 should change from available to occupied.	To be entered after manual execution	To be entered
IT_04_03	Slot Availability	Reflect vehicle exit release	Exit module is available	Complete a vehicle exit; open availability.	Vehicle: KA01AB1234; Slot A01	A01 should change from occupied to available.	To be entered after manual execution	To be entered
ST_04_01	Slot Availability	Verify availability before reservation	User is logged in; slots exist	View slots and choose an available slot for reservation.	A05 available	Only currently available slots should be selectable.	To be entered after manual execution	To be entered
ST_04_02	Slot Availability	Verify unavailable slot cannot be selected	A chosen slot is occupied/unavailable	Attempt to select it in a parking transaction.	A06 occupied	Selection or booking should be blocked.	To be entered after manual execution	To be entered

FR-05 - Vehicle Entry

SRS requirement: The system shall record vehicle entry time and allocate an available parking slot.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_05_01	Vehicle Entry	Record entry time	Entry module is accessible	Register a valid vehicle entry.	Vehicle: KA01AB1234; Current time	Entry record should contain the correct entry timestamp.	To be entered after manual execution	To be entered
UT_05_02	Vehicle Entry	Allocate an available slot on entry	An available slot exists	Enter a valid vehicle and complete entry.	Vehicle: KA01AB1234; Slot A01 available	System should allocate A01.	To be entered after manual execution	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_05_03	Vehicle Entry	Reject entry without vehicle details	Entry form is available	Submit without a vehicle number.	Vehicle: blank	Entry should be rejected with validation.	To be entered after manual execution	To be entered
IT_05_01	Vehicle Entry	Update slot status after entry	Vehicle and available slot exist	Complete vehicle entry; then view slot status.	Vehicle: KA01AB1234; A01	A01 should become occupied.	To be entered after manual execution	To be entered
IT_05_02	Vehicle Entry	Prevent allocation of an occupied slot	A01 is occupied	Attempt to allocate A01 during entry.	A01 occupied	System should select another available slot or reject the allocation.	To be entered after manual execution	To be entered
IT_05_03	Vehicle Entry	Store entry against the correct vehicle	Vehicle is registered	Complete entry and open vehicle/session record.	KA02CD5678; A02	The entry record should be linked to KA02CD5678.	To be entered after manual execution	To be entered
ST_05_01	Vehicle Entry	Complete vehicle entry from user input to allocation	Application and slot inventory are available	Enter vehicle information and submit entry.	KA03EF9012	Entry time should be recorded and an available slot allocated.	To be entered after manual execution	To be entered
ST_05_02	Vehicle Entry	Handle entry when no slots are available	All slots are occupied	Attempt to record a new vehicle entry.	All slots occupied	System should not allocate a slot and should clearly indicate no availability.	To be entered after manual execution	To be entered

FR-06 - Vehicle Exit

SRS requirement: The system shall record vehicle exit time and release the occupied parking slot.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_06_01	Vehicle Exit	Record vehicle exit time	Active parking session exists	Initiate vehicle exit and confirm.	KA01AB1234; A01	Exit timestamp should be recorded.	To be entered after manual execution	To be entered
UT_06_02	Vehicle Exit	Release occupied slot after exit	Vehicle occupies A01	Complete exit process.	Vehicle: KA01AB1234; A01	A01 should be released.	To be entered after manual execution	To be entered
UT_06_03	Vehicle Exit	Reject exit for vehicle without active session	No active session exists	Search/submit exit for a vehicle not currently parked.	KA09ZZ9999	Exit should be rejected or report no active session.	To be entered after manual execution	To be entered
IT_06_01	Vehicle Exit	Update slot availability after exit	Active session and availability module exist	Complete exit then open availability.	KA01AB1234; A01	A01 should become available.	To be entered after manual execution	To be entered
IT_06_02	Vehicle Exit	Link exit record to original entry record	Entry exists for vehicle	Complete exit and inspect session/history.	KA02CD5678	Entry and exit timestamps should belong to the same parking session.	To be entered after manual execution	To be entered
IT_06_03	Vehicle Exit	Calculate duration using entry and exit	Entry exists	Exit a vehicle after a known elapsed period.	Entry 10:00; Exit 12:00	Recorded duration should correspond to the elapsed parking time.	To be entered after manual execution	To be entered
ST_06_01	Vehicle Exit	Complete vehicle exit end-to-end	Active vehicle is parked	Find vehicle, process exit, verify slot release.	KA03EF9012; A03	Exit should complete, duration should be recorded and slot released.	To be entered after manual execution	To be entered
ST_06_02	Vehicle Exit	Prevent duplicate exit for same session	Session already closed	Attempt to process exit again.	KA03EF9012	System should prevent a second exit transaction.	To be entered after manual execution	To be entered

FR-07 - Parking Reservation

SRS requirement: The system shall allow users to reserve an available parking slot for a specified period.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_07_01	Parking Reservation	Create reservation for available slot	User is logged in; slot is available	Select slot and valid start/end period; reserve.	A01; 10-Sep-2026 10:00-12:00	Reservation should be created.	To be entered after manual execution	To be entered
UT_07_02	Parking Reservation	Reject reservation for occupied slot	A02 is occupied	Try to reserve A02.	A02	Reservation should be blocked.	To be entered after manual execution	To be entered
UT_07_03	Parking Reservation	Reject reservation with invalid period	Reservation form is available	Enter an end time earlier than start time.	12:00 to 10:00	Invalid time range should be rejected.	To be entered after manual execution	To be entered
IT_07_01	Parking Reservation	Update slot/reservation state after reservation	Reservation and availability modules are available	Reserve A03; reopen availability.	A03; 10-Sep-2026	Reserved slot/time should no longer be freely reservable for the same period.	To be entered after manual execution	To be entered
IT_07_02	Parking	Prevent	Existing	Attempt another	A04; 10:00-12:00	Overlapping	To be entered	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
	Reservation	overlapping reservation for same slot	reservation exists	reservation overlapping the same slot/time.	and 11:00-13:00	reservation should be blocked.	after manual execution	
IT_07_03	Parking Reservation	Associate reservation with correct user/vehicle	User and vehicle exist	Create reservation and open reservation details.	User Ravi; Vehicle KA01AB1234; A05	Correct user and vehicle should be recorded.	To be entered after manual execution	To be entered
ST_07_01	Parking Reservation	Complete reservation flow	Application is running; slot is available	Log in; choose slot; choose period; confirm reservation.	A06; 14:00-16:00	Confirmation should be generated with correct details.	To be entered after manual execution	To be entered
ST_07_02	Parking Reservation	Cancel/expire reservation and restore availability	A reservation exists and system supports cancellation/expiry	Cancel or let the reservation expire; check availability.	A06	Slot should become reservable according to configured rules.	To be entered after manual execution	To be entered

FR-08 - Parking Fee Calculation

SRS requirement: The system shall calculate parking fees based on parking duration and applicable rates.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_08_01	Parking Fee Calculation	Calculate fee for known duration	Rate is configured	Enter a valid parking duration.	2 hours; Rate ₹50/hour	Fee should be ₹100.	To be entered after manual execution	To be entered
UT_08_02	Parking Fee Calculation	Calculate fee for zero/invalid duration	Calculation module is available	Enter zero or invalid duration.	0 hours / -1 hours	System should reject invalid duration or apply defined minimum rule.	To be entered after manual execution	To be entered
UT_08_03	Parking Fee Calculation	Calculate fee for partial duration	Rate rule is known	Enter a partial-hour duration.	1.5 hours; Rate ₹50/hour	Fee should follow the configured rounding/pricing rule.	To be entered after manual execution	To be entered
IT_08_01	Parking Fee Calculation	Use actual entry and exit times for fee	Entry and exit records exist	Complete a session and calculate fee.	10:00 to 12:00; ₹50/hour	Fee should reflect the recorded 2-hour duration.	To be entered after manual execution	To be entered
IT_08_02	Parking Fee Calculation	Use applicable parking rate	Different rate configurations exist	Calculate fee for a slot/time with its applicable rate.	A01; ₹60/hour; 2 hours	Correct configured rate should be applied.	To be entered after manual execution	To be entered
IT_08_03	Parking Fee Calculation	Show fee before payment	Valid calculated fee exists	Proceed from completed parking session to payment.	Calculated fee ₹120	Payment screen should display the same payable amount.	To be entered after manual execution	To be entered
ST_08_01	Parking Fee Calculation	Verify complete fee calculation flow	Entry/exit and rate data are available	Complete a parking session and review final fee.	2-hour session; configured rate	Displayed fee should match duration and applicable rate.	To be entered after manual execution	To be entered
ST_08_02	Parking Fee Calculation	Prevent payment using altered amount	Calculated fee is ₹100	Attempt to alter payable amount before payment.	Original ₹100; attempted ₹10	System should preserve/validate the correct calculated amount.	To be entered after manual execution	To be entered

FR-09 - Payment Management

SRS requirement: The system shall allow users to pay the calculated parking fee through supported payment methods.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_09_01	Payment Management	Pay a valid parking fee	Valid unpaid fee exists	Select a supported payment method and submit valid test payment.	Fee ₹100; Valid test payment	Payment should succeed in the test environment.	To be entered after manual execution	To be entered
UT_09_02	Payment Management	Reject invalid payment data	Unpaid fee exists	Enter invalid/incomplete test payment data.	Invalid test payment data	Payment should be rejected with an error.	To be entered after manual execution	To be entered
UT_09_03	Payment Management	Display supported payment methods	Payment module is accessible	Open payment page.	Configured payment methods	Supported methods should be displayed.	To be entered after manual execution	To be entered
IT_09_01	Payment Management	Update payment status after successful payment	Booking/fee and payment modules are connected	Complete successful payment; reopen transaction.	Fee ₹100	Payment status should become paid/successful.	To be entered after manual execution	To be entered
IT_09_02	Payment Management	Do not mark payment paid after failure	Payment attempt can fail	Submit invalid test payment; view transaction.	Invalid payment	Status should remain unpaid/failed.	To be entered after manual execution	To be entered
IT_09_03	Payment Management	Prevent duplicate payment for same transaction	Transaction is already paid	Attempt to pay the same transaction again.	Transaction ID: T001	System should prevent duplicate charge or clearly handle it.	To be entered after manual execution	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST_09_01	Payment Management	Complete end-to-end payment flow	Valid parking transaction exists	Open fee, choose method, complete test payment, verify result.	Fee ₹150; valid test payment	Payment should complete and confirmation should be recorded.	To be entered after manual execution	To be entered
ST_09_02	Payment Management	Display final payment status to user	Payment has completed or failed	Open payment/transaction details.	Transaction T002	Correct current payment status should be visible.	To be entered after manual execution	To be entered

FR-10 - Parking History

SRS requirement: The system shall maintain records of previous parking sessions, including entry time, exit time, slot, and fee.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_10_01	Parking History	Store completed parking session	A vehicle completes entry and exit	Complete a parking session and save it.	KA01AB1234; A01; Entry 10:00; Exit 12:00	Completed session should be stored.	To be entered after manual execution	To be entered
UT_10_02	Parking History	Display entry and exit times	History contains a completed session	Open parking history.	Entry 10:00; Exit 12:00	Both timestamps should be displayed correctly.	To be entered after manual execution	To be entered
UT_10_03	Parking History	Display slot in history	History contains a completed session	Open session details.	Slot A01	A01 should be shown.	To be entered after manual execution	To be entered
IT_10_01	Parking History	Display fee in history	Payment/fee exists	Open history after payment.	Fee ₹100	Correct fee should be shown.	To be entered after manual execution	To be entered
IT_10_02	Parking History	Associate history with correct vehicle	Multiple vehicles exist	Open history for one vehicle/user.	KA02CD5678	Only the correct related records should be shown/identified.	To be entered after manual execution	To be entered
IT_10_03	Parking History	Preserve history after slot is released	Completed session released its slot	Reopen history after exit.	A03 released	Historical session should remain even though the slot is available.	To be entered after manual execution	To be entered
ST_10_01	Parking History	Verify complete parking-history record	Completed session is available	Review history entry from end to end.	Vehicle, entry, exit, slot, fee	All required historical fields should be present and accurate.	To be entered after manual execution	To be entered
ST_10_02	Parking History	Handle user with no previous sessions	User has never parked	Open parking history.	No sessions	System should show an appropriate empty-history message.	To be entered after manual execution	To be entered

FR-11 - Notifications

SRS requirement: The system shall notify users about reservation confirmation, parking expiry, and payment status.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_11_01	Notifications	Send reservation confirmation notification	Valid reservation is created	Create a reservation and check notifications.	Reservation A01	Confirmation notification should be generated.	To be entered after manual execution	To be entered
UT_11_02	Notifications	Notify about parking expiry	Reservation/session has an expiry point	Reach or simulate the expiry event.	Expiry: 16:00	Expiry notification should be generated.	To be entered after manual execution	To be entered
UT_11_03	Notifications	Notify payment status	Payment succeeds or fails	Complete a payment attempt and check notifications.	Payment success/failure	Corresponding status notification should be generated.	To be entered after manual execution	To be entered
IT_11_01	Notifications	Trigger notification after reservation event	Reservation and notification modules are connected	Create reservation; inspect notification center/log.	Reservation R001	Reservation event should trigger the correct notification.	To be entered after manual execution	To be entered
IT_11_02	Notifications	Trigger notification after payment event	Payment and notification modules are connected	Complete payment; inspect notification state.	Transaction T001	Payment status should be reflected in notification.	To be entered after manual execution	To be entered
IT_11_03	Notifications	Show notification to the correct user	Two user accounts exist	Create an event for User A; inspect User B's notifications.	User A reservation	User B should not receive User A's private notification.	To be entered after manual execution	To be entered
ST_11_01	Notifications	Verify complete reservation confirmation notification	User completes reservation	Create reservation and view notification.	R002	Correct slot, period and confirmation status should be shown.	To be entered after manual execution	To be entered
ST_11_02	Notifications	Verify notification	Notification exists	Refresh	Notification N001	Notification	To be entered	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
		state after repeated refresh		notification center multiple times.		should remain consistent and not duplicate unexpectedly.	after manual execution	

FR-12 - Admin Dashboard

SRS requirement: The administrator shall be able to view parking occupancy, active vehicles, reservations, and revenue.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_12_01	Admin Dashboard	Display parking occupancy	Admin is logged in; slot data exists	Open dashboard occupancy section.	5 occupied of 20	Displayed occupancy should match current slot data.	To be entered after manual execution	To be entered
UT_12_02	Admin Dashboard	Display active vehicles	Active parking sessions exist	Open dashboard.	3 active vehicles	Dashboard should display the correct count/list.	To be entered after manual execution	To be entered
UT_12_03	Admin Dashboard	Display reservations	Reservations exist	Open dashboard reservation section.	4 active reservations	Correct reservation information/count should be shown.	To be entered after manual execution	To be entered
UT_12_04	Admin Dashboard	Display revenue	Completed paid transactions exist	Open dashboard revenue section.	Transactions totaling ₹500	Displayed revenue should match the underlying paid transactions.	To be entered after manual execution	To be entered
IT_12_01	Admin Dashboard	Update occupancy after vehicle entry/exit	Dashboard and entry/exit modules are available	Record an entry and then exit; refresh dashboard.	A01 entry then exit	Occupancy should update correctly.	To be entered after manual execution	To be entered
IT_12_02	Admin Dashboard	Update revenue after successful payment	Payment and dashboard are connected	Complete a valid payment; refresh dashboard.	Payment ₹100	Revenue should reflect the successful payment.	To be entered after manual execution	To be entered
ST_12_01	Admin Dashboard	Verify dashboard shows all required admin metrics	Admin is logged in; data exists	Open dashboard and inspect occupancy, active vehicles, reservations and revenue.	Representative test data	All four required information areas should be visible and accurate.	To be entered after manual execution	To be entered
ST_12_02	Admin Dashboard	Prevent normal user access to admin dashboard	Normal user is logged in	Attempt to open admin dashboard.	Normal user session	Access should be denied.	To be entered after manual execution	To be entered

FR-13 - Search Vehicle

SRS requirement: The administrator shall be able to search for vehicles using vehicle number or other relevant details.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_13_01	Search Vehicle	Search by exact vehicle number	Vehicle record exists	Enter exact vehicle number and search.	KA01AB1234	Matching vehicle should be displayed.	To be entered after manual execution	To be entered
UT_13_02	Search Vehicle	Search by non-existing number	No matching record exists	Enter unknown vehicle number and search.	KA99ZZ9999	No-result message should be displayed.	To be entered after manual execution	To be entered
UT_13_03	Search Vehicle	Search with partial/relevant detail	Search supports relevant details	Enter supported partial detail.	KA01	Matching records should be returned if partial search is supported.	To be entered after manual execution	To be entered
IT_13_01	Search Vehicle	Show current parking state in vehicle search	Vehicle has an active session	Search for active vehicle.	KA02CD5678	Search should show relevant current record/state.	To be entered after manual execution	To be entered
IT_13_02	Search Vehicle	Show vehicle information from registration data	Vehicle is registered	Search the vehicle and inspect fields.	KA03EF9012	Vehicle number/type/owner information should match stored data.	To be entered after manual execution	To be entered
IT_13_03	Search Vehicle	Respect admin authorization	Admin and normal-user accounts exist	Attempt vehicle search as each role.	Admin vs normal user	Behavior should follow the SRS-defined administrator search capability.	To be entered after manual execution	To be entered
ST_13_01	Search Vehicle	Complete admin vehicle search flow	Admin is logged in	Search by vehicle number and inspect result.	KA04GH3456	Correct vehicle record should be returned.	To be entered after manual execution	To be entered
ST_13_02	Search Vehicle	Handle special/invalid search input safely	Search field is available	Enter empty or invalid special input and search.	Empty; !!!	System should handle the input safely without errors or unintended	To be entered after manual execution	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
						access.		

FR-14 - Reports

SRS requirement: The system shall generate reports related to parking usage, revenue, and vehicle activity.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_14_01	Reports	Generate parking usage report	Report data exists	Select parking usage report and generate.	10 sessions; September 2026	Usage report should be generated.	To be entered after manual execution	To be entered
UT_14_02	Reports	Generate revenue report	Paid transactions exist	Generate revenue report for a period.	₹100, ₹150, ₹250	Revenue report should reflect the underlying payments.	To be entered after manual execution	To be entered
UT_14_03	Reports	Generate vehicle activity report	Vehicle activity exists	Generate vehicle activity report.	KA01AB1234; KA02CD5678	Activity data should be reported correctly.	To be entered after manual execution	To be entered
IT_14_01	Reports	Use parking history in reports	History records exist	Generate usage report and compare with history.	5 completed sessions	Report totals/details should align with history.	To be entered after manual execution	To be entered
IT_14_02	Reports	Use payment data in revenue reports	Payment records exist	Generate revenue report and compare with payments.	3 paid transactions	Report should reflect successful payment records.	To be entered after manual execution	To be entered
IT_14_03	Reports	Filter report by a valid period	Report supports period selection	Generate report for a specific date range.	01-Sep-2026 to 10-Sep-2026	Only data within the selected range should be included.	To be entered after manual execution	To be entered
ST_14_01	Reports	Generate complete reporting output	Admin has access and data exists	Generate usage, revenue and vehicle activity reports.	September 2026 dataset	Required reports should generate successfully and contain coherent data.	To be entered after manual execution	To be entered
ST_14_02	Reports	Handle empty report period	No records exist for selected period	Generate a report for a period with no data.	01-Jan-2025 to 02-Jan-2025	System should show a clear empty/no-data result rather than incorrect totals.	To be entered after manual execution	To be entered

5. Non-Functional Test Cases

NFR-01 - Performance

Priority: High. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-01-01	Performance	Measure normal request response time	Application is running under normal load	Open dashboard, search vehicle, and view availability; measure response.	Normal test dataset	Requests should respond within the project's defined acceptable time.	To be entered after manual execution	To be entered
NFR-01-02	Performance	Measure search response under normal load	Vehicle dataset is populated	Run multiple vehicle searches and record response time.	20 representative searches	Response should remain within acceptable normal-operating limits.	To be entered after manual execution	To be entered
NFR-01-03	Performance	Measure booking response time	Slots and user data exist	Create a reservation and measure time from submit to confirmation.	Reservation R001	Confirmation should return within acceptable time.	To be entered after manual execution	To be entered
NFR-01-04	Performance	Measure dashboard loading	Admin dashboard has data	Load dashboard and record response time.	Representative occupancy/revenue data	Dashboard should load within acceptable time.	To be entered after manual execution	To be entered

NFR-02 - Security

Priority: High. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-02-01	Security	Reject	Normal user	Attempt direct	Normal user	Unauthorized	To be entered	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
		unauthorized admin access	account exists	access to admin dashboard/slot management.	session	access should be denied.	after manual execution	
NFR-02-02	Security	Protect login with incorrect credentials	Valid account exists	Try incorrect password repeatedly in a controlled test.	Wrong password	System should not grant access.	To be entered after manual execution	To be entered
NFR-02-03	Security	Verify protected session after logout	User is logged in	Log out and try to access a protected page using the old session/browser state.	Logged-out session	Protected content should not remain accessible.	To be entered after manual execution	To be entered
NFR-02-04	Security	Verify sensitive data is not exposed in normal UI	Accounts/payment test data exist	Inspect ordinary user/admin pages for exposed credentials or unnecessary sensitive values.	Test account data	Credentials and sensitive values should not be unnecessarily exposed.	To be entered after manual execution	To be entered

NFR-03 - Reliability

Priority: High. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-03-01	Reliability	Verify records persist after reload	Records exist	Create/update a vehicle or slot; reload application.	Vehicle KA01AB1234	Saved record should remain accurate.	To be entered after manual execution	To be entered
NFR-03-02	Reliability	Verify records persist after logout/login	User has saved data	Save vehicle; log out; log back in; inspect data.	KA02CD5678	Data should persist without loss.	To be entered after manual execution	To be entered
NFR-03-03	Reliability	Verify slot state remains consistent	Vehicle occupies a slot	Refresh/reopen availability and booking views.	A01 occupied	Slot state should remain consistent.	To be entered after manual execution	To be entered
NFR-03-04	Reliability	Verify duplicate operation does not corrupt records	Existing transaction exists	Repeat an action such as booking/payment submission where duplicates are not allowed.	Transaction T001	Records should remain consistent without unintended duplicate state.	To be entered after manual execution	To be entered

NFR-04 - Usability

Priority: Medium. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-04-01	Usability	Check clarity of navigation	Application is available	Navigate through user registration, vehicle and parking modules.	Representative user flow	Navigation should be understandable and consistent.	To be entered after manual execution	To be entered
NFR-04-02	Usability	Check form error messages	Forms are available	Submit invalid registration/vehicle/booking data.	Invalid input set	Messages should clearly identify what needs correction.	To be entered after manual execution	To be entered
NFR-04-03	Usability	Check admin dashboard readability	Admin is logged in	Review dashboard sections and labels.	Representative dashboard data	Information should be presented clearly and intuitively.	To be entered after manual execution	To be entered
NFR-04-04	Usability	Check consistency of status labels	Reservations/payments/slots exist	Compare labels across related screens.	Available, occupied, paid, pending	Status terminology should be clear and consistent.	To be entered after manual execution	To be entered

NFR-05 - Availability

Priority: High. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-05-01	Availability	Verify application opens when services are available	Application services are running	Open the application in supported browser.	Normal environment	Application should be accessible.	To be entered after manual execution	To be entered
NFR-05-02	Availability	Verify availability page can be reached during normal	System is running	Open slot availability repeatedly.	Normal usage	Availability information should remain accessible.	To be entered after manual execution	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
		operations						
NFR-05-03	Availability	Verify user can return to system after temporary interruption	System is restarted in a controlled environment	Restart service; reopen application.	Controlled restart	Application should become accessible again and retained data should remain intact.	To be entered after manual execution	To be entered
NFR-05-04	Availability	Verify key parking functions remain available during normal operating period	System is operational	Use entry, exit and availability functions in sequence.	Normal operating scenario	Required parking services should remain usable.	To be entered after manual execution	To be entered

NFR-06 - Scalability

Priority: Medium. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-06-01	Scalability	Increase vehicle records and verify core functions	Test database can accept larger dataset	Add many representative vehicle records and run search.	1,000 test vehicle records	Core functions should continue to operate.	To be entered after manual execution	To be entered
NFR-06-02	Scalability	Increase parking-slot records	Admin can add slots	Populate a larger slot inventory and open availability.	500 test slots	Availability should remain usable and accurate.	To be entered after manual execution	To be entered
NFR-06-03	Scalability	Increase user records	Registration/admin data setup is available	Populate many test user records and perform login/search operations.	1,000 test users	Core user operations should continue to work.	To be entered after manual execution	To be entered
NFR-06-04	Scalability	Verify report generation with larger dataset	Large test dataset exists	Generate usage/revenue report over the larger dataset.	Large September 2026 dataset	Reports should complete without incorrect totals or system failure.	To be entered after manual execution	To be entered

NFR-07 - Maintainability

Priority: Medium. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-07-01	Maintainability	Verify parking rate can be modified	Admin/settings access exists per implementation	Change a configured parking rate; perform a fee calculation.	₹50/hour -> ₹60/hour	New rate should be used in subsequent calculations.	To be entered after manual execution	To be entered
NFR-07-02	Maintainability	Verify slot configuration can be updated	Admin can modify slot settings	Change slot configuration and inspect availability.	A01 configuration update	Updated configuration should be reflected without unnecessary code changes to the user flow.	To be entered after manual execution	To be entered
NFR-07-03	Maintainability	Verify system settings are manageable through intended configuration	Configuration mechanism exists	Update a supported setting and reload relevant module.	Configured test setting	New setting should be used consistently.	To be entered after manual execution	To be entered
NFR-07-04	Maintainability	Verify changes do not alter unrelated records	Existing vehicle/session data exists	Change a supported rate/slot setting and inspect unrelated records.	Existing test data	Unrelated records should remain unchanged.	To be entered after manual execution	To be entered

NFR-08 - Data Integrity

Priority: High. These tests verify the quality attribute stated in the SRS.

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-08-01	Data Integrity	Prevent one slot from being assigned to two vehicles simultaneously	A01 is available	Attempt to allocate A01 to two vehicles at the same time.	KA01AB1234 and KA02CD5678	Only one vehicle should receive A01.	To be entered after manual execution	To be entered
NFR-08-02	Data Integrity	Keep slot state consistent with active parking session	Active vehicle uses A02	Compare vehicle session and slot state.	KA03EF9012; A02	Active session should correspond to occupied A02.	To be entered after manual execution	To be entered
NFR-08-03	Data Integrity	Prevent	Reservation	Create	A03; overlapping	Second	To be entered	To be entered

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
		duplicate reservation of same slot and overlapping period	R001 exists	overlapping reservation for same slot.	period	conflicting reservation should be rejected.	after manual execution	
NFR-08-04	Data Integrity	Preserve correct fee and session linkage	Completed session and payment exist	Inspect history, fee and payment records together.	Session S001; Fee ₹100	Related records should refer to the same vehicle/session and correct amount.	To be entered after manual execution	To be entered

6. Manual Test Execution Summary

SRS ID	Total Cases	Unit	Integration	System
FR-01	8	3	3	2
FR-02	8	3	3	2
FR-03	8	4	2	2
FR-04	8	3	3	2
FR-05	8	3	3	2
FR-06	8	3	3	2
FR-07	8	3	3	2
FR-08	8	3	3	2
FR-09	8	3	3	2
FR-10	8	3	3	2
FR-11	8	3	3	2
FR-12	8	4	2	2
FR-13	8	3	3	2
FR-14	8	3	3	2

Functional test cases planned: 112 (8 per FR × 14 FRs).

Non-functional test cases planned: 32

7. Defect / Observation Log

Defect ID	SRS ID	Test Case ID	Description	Severity	Status

8. Test Completion Rules

- Execute every planned test case manually.
- Enter the actual observed behavior in Actual Result.
- Mark Test Result as Pass only when Actual Result matches Expected Results; otherwise mark Fail.
- Record failed cases and defects in the defect/observation log.
- After fixing defects, re-run the affected cases and update the result.
- Use this document together with the SRS so every test can be traced back to a requirement.

9. Source Alignment Note

This document uses the terminology and functional/non-functional requirement structure from the supplied Vehicle Parking System SRS. Where the SRS does not specify exact implementation details, the test cases use generic test data and describe the expected behavior at the requirement level rather than inventing product-specific UI labels.